

JOON KIM

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EDUCATION

University of California, Berkeley

May 2027 (Expected)

B.S. Electrical Engineering & Computer Science, GPA: 4.00/4.00

Current: CS274(Computational Geometry), CS194-198(Network Theory), CS294-268(LEAN4)

TCS: CS270(Graduate Algorithms), CS294-180(Partition Functions, **A+**), CS174(Randomized Algorithms), CS170(Algorithms, **A+**), CS177(Algorithmic Economics), CS294-284(Constraint Satisfaction Problems)

Computational: CS194-302(Computational Immunology, **A+**), EECS249B(Cyber-Physical Systems), CS176(Computational Biology), CS188(AI/RL)

RESEARCH INTERESTS

Algorithms; MCMC and Partition Functions; Combinatorial Optimization

RESEARCH & PRESENTATIONS

[Foundational Theory]

MCMC Research

Jan. 2026 - Current

Advised by Prof. Alistair Sinclair

Berkeley, CA

- Studying the mixing time lower bound of lattice triangulation Glauber dynamics for $\lambda > 1$ in [CMSS15]
- Attempting to improve T_{mix} from $e^{\Omega((m+n))}$ to $e^{\Omega((m+n)mn)}$ by bounding the conductance of the flip graph

Maximum Weighted Matching Research

Dec. 2025 - Current

Advised by Prof. Satish Rao

Berkeley, CA

- Extending Multiplicative Auction by [ZH25] to fully dynamic matching; identified monotonicity as a barrier
- Modifying the auction algorithm, taking inspiration from real-life market mechanisms and human behavior

Sampling Spanning Trees via Entropic Independence

Oct. 2025 - Dec. 2025

CS294-180(Partition Functions) Final Report & Student Lecture

- Synthesized a 6-page report for sublinear time sampling of spanning trees in $O(k \log^2 n)$ time, based on [ALV22]
- Delivered half of a 70-minute technical lecture to a graduate-level audience, motivating and explaining techniques such as marginal overestimates, isotropic transformation, MCMC, and connections to Up-Down walks

[Interdisciplinary Studies]

Lawrence Berkeley Laboratory - Perlmutter Group

Sep. 2025 - Current

Advised by Prof. Xiaosheng Huang

Berkeley, CA

- Modeling the Carousel Lens using the multi-GPU GIGA-Lens ML pipeline to constrain cosmology
- Implemented high-dimensional Stochastic Variational Inference with Gaussian Mixtures and Dirichlet Processes

Stochastic Modeling of Drug-Induced Resistance

Sep. 2025 - Dec. 2025

CS194-302(Computational Immunology) Final Report & Presentation

- Designed an explainable computational model for cancer drug resistance using Hill-function modulated CTMC
- Delivered half of a 15-minute conference-style presentation to a graduate-level audience
- Potential submission to workshop or conference with minor revisions

Cohort Studies and MrVI

Nov. 2025

CS194-302(Computational Immunology) Student Lecture

- Delivered a third of a 3-hour lecture on cohort studies in immunology and MrVI as one emerging solution
- Focused on building mathematical intuition of MrVI, how it differs from scVI, and its philosophical implications

Greedy, ILP, and Hybrid Methods for Phylogeny

Apr. 2025

CS176(Computational Biology) Literature Review

- Synthesized a 5-page technical review on Casasiopeia, a phylogeny framework with ILP and Greedy heuristics
- Identified motivation for the study, its main technical contributions, potential applications, and limitations.

Berkeley AI Research - C.H.E.N. Lab

Jul. 2024 - Feb. 2025

Advised by Prof. Irene Chen

Berkeley, CA

- Designed zero-shot LLM pseudo-label pipeline to improve semi-supervised learning accuracy
- Investigated LLM agents for image labeling such as CLIP and VIT, showed results on CIFAR-100
- Worked on RadQA dataset; implemented FixMatch and a new method on a non-inference task for comparison

JLK Group

Feb. 2024 - May 2024

Research Intern, advised by Dr. Wi-Sun Ryu

Seoul, South Korea

- Developed Federated Learning models reaching near identical performance to commercially deployed U-Net models
- Collaborated with four M.D. professionals to investigate the use of Federated Learning in medical imaging

[Security & Safety]

StSTL Verification via Scenario Optimization

Sep. 2025 - Current

EECS249B(Cyber-Physical Systems) Final Project, advised by Prof. Pierluigi Nuzzo

- Investigated distribution-free probabilistic guarantees in stochastic environments for contract-based design
- Applied Scenario Optimization (SO) to a simple autonomous vehicle example for static and runtime verification
- Currently attempting to apply SO in distribution-free StSTL verification on the optimization level

University of Florida REU

May 2025 - Aug. 2025

Advised by Prof. Sandip Ray

Gainesville, FL

- Explored network-level bit flipping attack and defense on 5G Connected and Automated Vehicles (CAV)
- Verified feasibility of bit-flipping; proposed a keystream-based shuffling that drastically lowers attack success rate

Keimyung University

Feb. 2023 - Jul. 2024

Advised by Prof. Sejin Park

Daegu, South Korea

- Proposed a randomized masking Federated Learning algorithm as an obfuscation against Deep Leakage for images
- Designed experiments to compare performance-privacy trade-offs amongst SOTA defense algorithms

TEACHING EXPERIENCE

CS70 Course Staff

Jan. 2026 - Current

8-hour Undergraduate Student Instructor

Berkeley, CA

- 4 hours of grading and 4 hours of discussion section assistance and OH for CS70 (Discrete Math and Probability)

Computer Science Mentors

Aug. 2025 - Dec. 2025

Junior Mentor

Berkeley, CA

- Led a group of three undergrads in biweekly hour-long review sessions for CS70 (Discrete Math and Probability)

PUBLICATIONS

J. Kim, C. Duan, S. Ray. "Bit-Flipping Attack Exploration and Countermeasure in 5G Network." *MASS 2025*.

J. Kim, H. Lee, W. Ryu, et al. "In-Silo Federated Learning vs. Centralized Learning for Segmenting Ischemic Brain Lesions." *Intelligence-Based Medicine*.

J. Kim, S. Park. "Random Gradient Masking as a Defensive Measure to Deep Leakage in Federated Learning." *ArXiv*.

C. Duan, **J. Kim**, S. Ray. "Network-Level Bit-Flipping Attacks on CACC in 5G." *Submitted to VTC2026-Spring*.